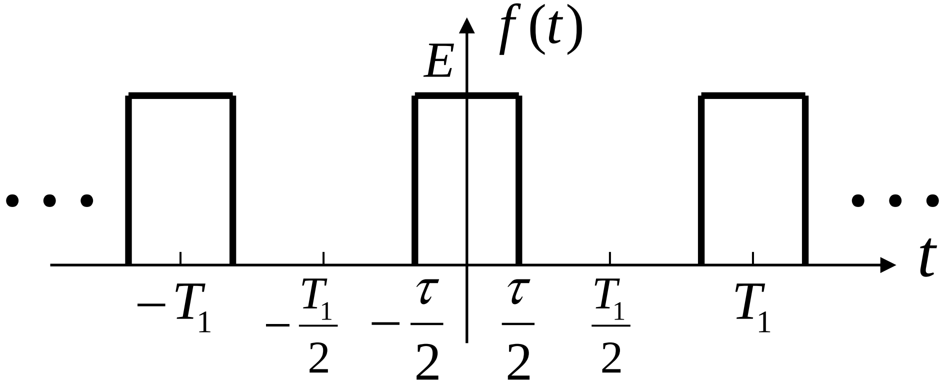
Review of Last Lecture

3.4 Fourier Transform

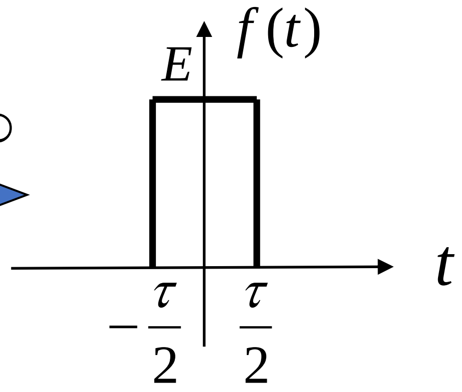
3.5 Fourier Transform of Typical Aperiodic Signals

3.6 Fourier Transform of Impulse and Step Functions

3.4 Fourier Transform



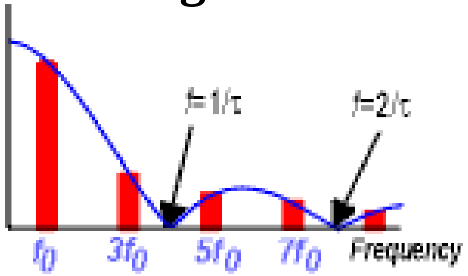
$T_1 \to \infty$



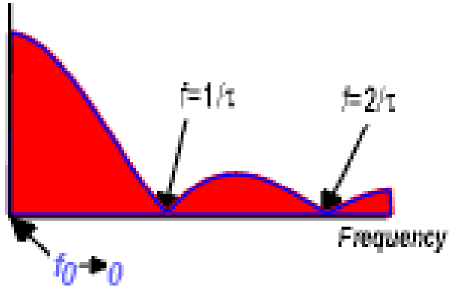
$T_1 \uparrow \quad \Rightarrow \quad \omega_1 = \frac{2\pi}{T_1} \downarrow \quad \Rightarrow \quad$ Spectral line spacing (proportional to ω_1) $\downarrow$

$T_1 \to \infty \quad \Rightarrow \quad \omega_1 = \frac{2\pi}{T_1} \to 0 \quad \Rightarrow \quad$ Spectral line spacing $\to 0$

Discrete Spectrum of Periodic Signals



Continuous Spectrum of Aperiodic Signals



Due to $T_1 \to \infty$, $$F_n = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)e^{-jn\omega_1 t}dt \to 0$$

Spectral Density Function: $$\lim_{T_1\to\infty} F_n T_1 = \lim_{T_1\to\infty}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)e^{-jn\omega_1 t}dt$$

When $T_1 \to \infty$, discrete frequency $n\omega_1 \to$ continuous frequency ω.

Then $$\lim_{T_1\to\infty} F_n T_1 = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

Let **F(ω)** denote the **Fourier Transform of aperiodic signal *f(t)***.

$$\boxed{F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt}$$

Inverse Fourier Transform

$$\boxed{f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega}$$

$$F(\omega) = |F(\omega)|e^{j\varphi(\omega)}$$

$|F(\omega)|$: Amplitude Spectrum

$\varphi(\omega)$: Phase Spectrum

Physical Meaning of Fourier Transform

It can be proved,

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)e^{j\omega t}\,d\omega=\int_{0}^{\infty}\underbrace{\frac{|F(\omega)|}{\pi}d\omega}_{\text{Amplitude}}\cos\left[\omega t+\varphi(\omega)\right]=\lim_{\Delta\omega\to 0}\sum_{n=0}^{\infty}\underbrace{\frac{|F(n\Delta\omega)|}{\pi}\Delta\omega}_{\text{Amplitude}}\cos\left[n\Delta\omega t+\varphi(n\Delta\omega)\right]$$

$f(t)$ is the sum of infinitely many continuous cosine signals with infinitesimal amplitude, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ covering frequencies from 0 to ∞.

Aperiodic signals can be expressed as a **weighted integral of sinusoidal signals**, containing all **continuous frequency components from zero to infinitely high frequencies**.

For energy-limited signals (e.g., single-pulse signals), the amplitude of each frequency component, $\left(\frac{1}{\pi}|F(\omega)|d\omega\right)$ tends to be infinitely small. Thus, the spectrum can no longer be represented by amplitude, but by the **spectral density function** instead.

Sufficient but Not Necessary Condition for the Existence of Fourier Transform

$$\boxed{\int_{-\infty}^{\infty} \left| f(t) \right| \mathrm{d}t < \infty \quad \text{i.e., } f(\mathrm{t}) \text{ is absolutely integrable.}}$$

Proof: $\int_{-\infty}^{\infty} \left| f(t) e^{-j\omega t} \right| \mathrm{d}t = \int_{-\infty}^{\infty} \left| f(t) \right| \mathrm{d}t < \infty$

Most **energy signals** satisfy this condition.

If a signal does not satisfy the above condition, Fourier transform may still exist, and may be conducted by introducing the concept of **impulse function δ**(t).

Relationship Between Fourier Series & Transform

Feature	Fourier Series	Fourier Transform
Expression	$f(t)=\sum_{n=-\infty}^{+\infty} F_n e^{jn\omega_1 t}$	$f(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega)e^{j\omega t}d\omega$
Spectrum Type	Discrete Spectrum $F_n=\frac{1}{T_1}\int_{-T_1/2}^{T_1/2} f(t)e^{-jn\omega_1 t}dt$ $F_n=\lvert F_n\rvert e^{j\varphi_n}$	Continuous Spectrum (**Spectral Density**) $F(\omega)=\int_{-\infty}^{+\infty} f(t)e^{-j\omega t}dt$ $F(\omega)=\lvert F(\omega)\rvert e^{j\varphi(\omega)}$
Existence Condition	Dirichlet conditions within one period $\int_0^T \lvert f(t)\rvert \mathrm{d}t<\infty$	Dirichlet conditions over (-∞, ∞) $\int_{-\infty}^{\infty} \lvert f(t)\rvert \mathrm{d}t<\infty$

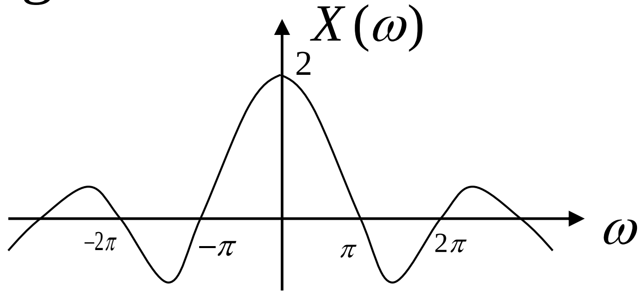
Common Functions and Their Fourier Transforms

Name	Time Function	Spectral Function
Unit Impulse	$\delta(t)$	1
Unit Step	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$
Signum Function	$\mathrm{sgn}(t)=u(t)-u(-t)$	$\frac{2}{j\omega}$
Unit DC Signal	1	$2\pi\delta(\omega)$
Single-sided Exponential Signal	$e^{-\alpha t}u(t)\ (\alpha>0)$	$\frac{1}{\alpha+j\omega}$
Odd-symmetric Two-sided Exponential Signal	$e^{-\alpha t}u(t)-e^{\alpha t}u(-t)\ (\alpha>0)$	$\frac{-2j\omega}{\alpha^2+\omega^2}$
Even-symmetric Two-sided Exponential Signal	$e^{-\alpha\lvert t\rvert}\ (\alpha>0)$	$\frac{2\alpha}{\alpha^2+\omega^2}$
Rectangular Pulse (Gate Function)	$E\left[u\left(t+\frac{\tau}{2}\right)-u\left(t-\frac{\tau}{2}\right)\right]$	$\mathrm{E}\tau\mathrm{Sa}\left(\frac{\omega\tau}{2}\right)$

单选题 1分

设置

Given that a signal $x(t)$ has a Fourier transform of the Sa function as shown in the figure, the time-domain expression of this signal is



A $x(t) = u(t+1) - u(t-1)$

B $x(t) = 2[u(t+1) - u(t-1)]$

C $x(t) = u(t+1/2) - u(t-1/2)$

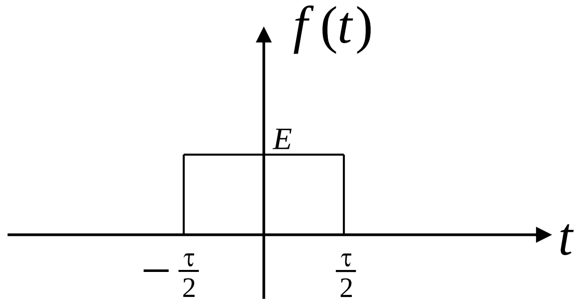
D $x(t) = 2[u(t+1/2) - u(t-1/2)]$

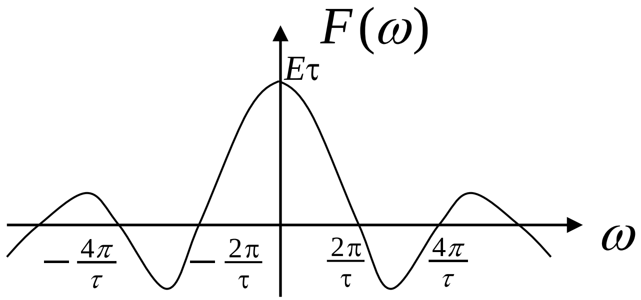
提交

3.5 Fourier Transform of Typical Aperiodic Signals

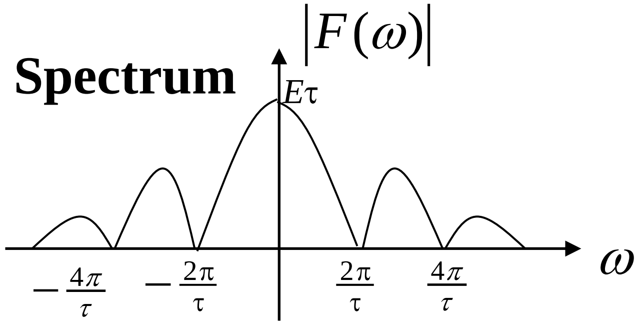
Rectangular Pulse Signal (Gate Function)

$$\boxed{E[u(t+\frac{\tau}{2})-u(t-\frac{\tau}{2})] \xleftrightarrow{FT} E\tau \mathrm{Sa}(\omega\frac{\tau}{2})}$$

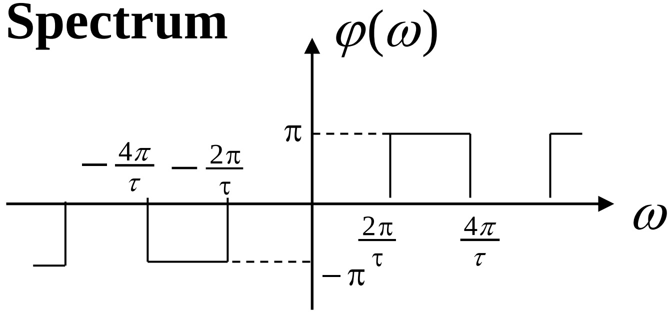




Amplitude Spectrum



Phase Spectrum



Equivalent Bandwidth

It can be seen from the above spectrum diagram of the rectangular wave that the signal energy is mainly concentrated within the frequency range $0 \sim 2\pi/\tau$. This range, $B\omega$ is called the **equivalent bandwidth** of the rectangular wave.

$$\boxed{B_\omega = \frac{2\pi}{\tau}\ \text{(rad/s) or } \frac{1}{\tau}\ \text{(Hz)}}$$

Contents of This Lecture

3.7 Basic Properties of Fourier Transform

Learning Objectives of This Lecture

Master the application of various properties of Fourier Transform to calculate the Fourier Transform of complex signals (**Important!**).

3.7 Basic Properties of Fourier Transform

- A time function and its spectral function (if exists) have a unique relationship.
- The basic properties of Fourier Transform allow us to derive the Fourier Transform of complex-time signals from that of typical signals.
- Conversely, we can infer the time-domain changes from frequency-domain operations.
- **Notes:**
  - Dirichlet's condition is a sufficient but not necessary condition.
  - For any given signal $\boldsymbol{f(t)}$, regardless of whether it satisfies the absolute integrability condition, Fourier analysis theory does not restrict the use of any method to obtain a formal transform $\boldsymbol{F(\omega)}$. However, some transforms may lack physical meaning.
  - Engineering practice shows:
    - Fourier Transform exists for most **energy signals**;
    - Generalized Fourier Transform exists for most **power signals** (e.g., step signal, signum signal);
    - Generalized Fourier Transform may not exist for non-power, non-energy signals.

Energy signal: $\int_{-\infty}^{\infty} |f(t)|^2 \, \mathrm{d}t < \infty$ Power signal: $\lim_{T_1 \to \infty} \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} |f(t)|^2 \, \mathrm{d}t < \infty$

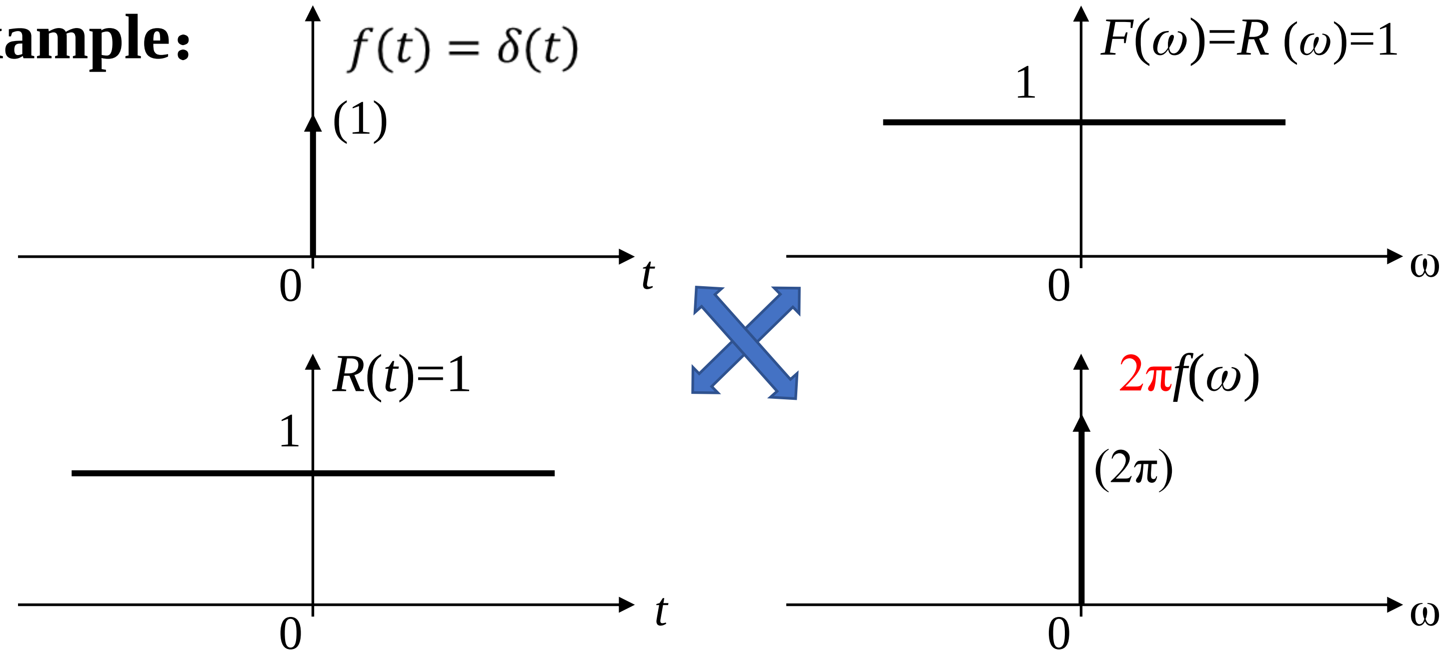
3.7.1 Linearity

if $\mathrm{F}[f_1(t)] = F_1(\omega),\ \mathrm{F}[f_2(t)] = F_2(\omega),$

then $\mathrm{F}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(\omega) + a_2 F_2(\omega)$

3.7.2 Symmetry

Example:



3.7 Basic Properties of Fourier Transform

Using the symmetry of Fourier Transform, the problem of finding the inverse Fourier Transform can be converted into finding the Fourier Transform.

if $\mathcal{F}[f(t)] = F(\omega)$ **then** $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

$$f(t) \leftrightarrow F(\omega)$$

$$F(t) \leftrightarrow 2\pi f(\omega) \quad (f \text{ is an even function})$$

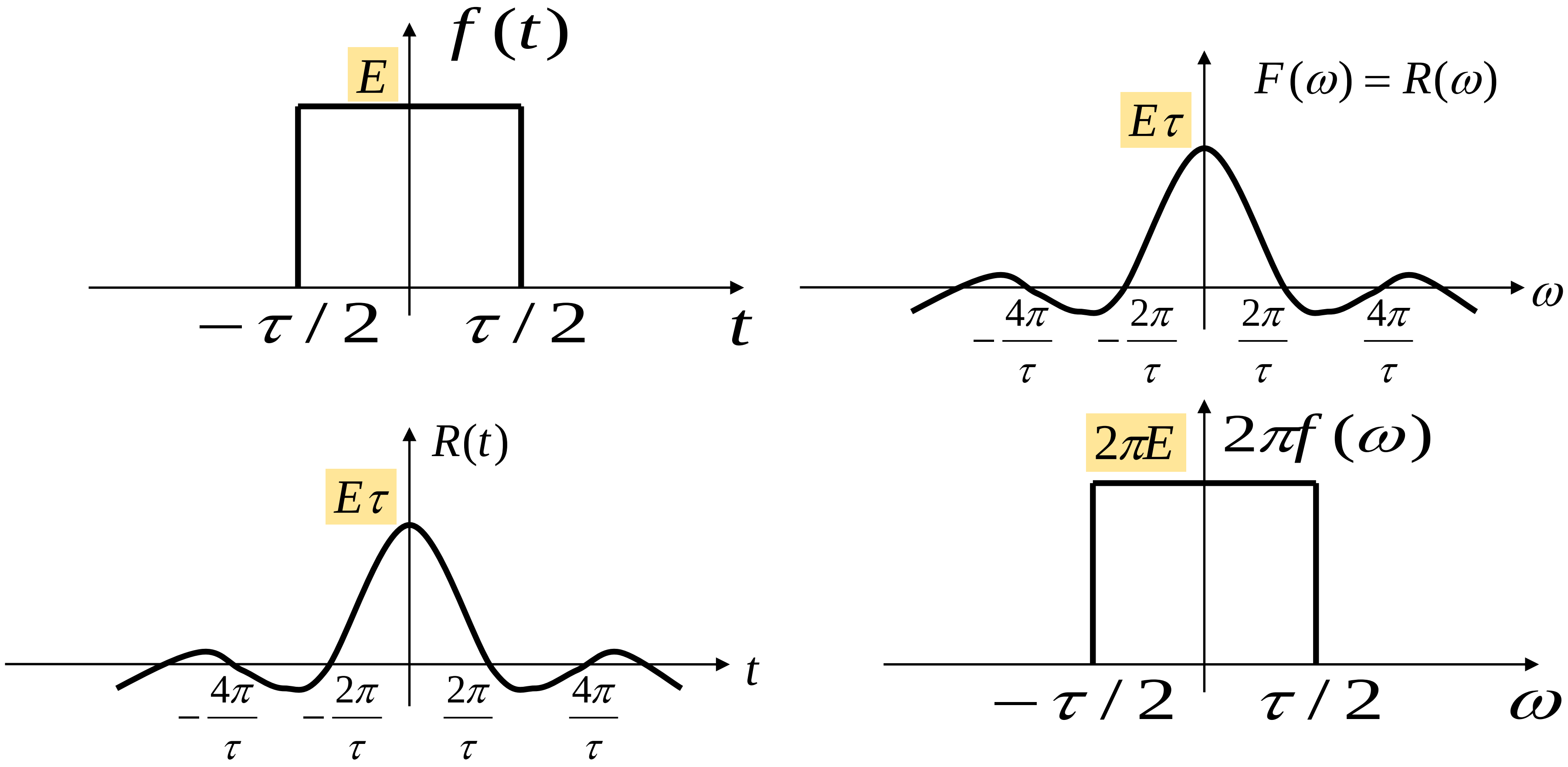
$$F(t) \leftrightarrow -2\pi f(\omega) \quad (f \text{ is an odd function})$$

$$f(t) \leftrightarrow F(\omega)$$

$$\frac{F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an even function})$$

$$\frac{-F(t)}{2\pi} \leftrightarrow f(\omega) \quad (f \text{ is an odd function})$$

Symmetry Property Example: Gate Function & Sampling Function



$$f(t)=E[u(t+\frac{\tau}{2})-u(t-\frac{\tau}{2})]\rightarrow R(\omega)=E\tau\mathrm{Sa}(\frac{\omega\tau}{2})$$

$$R(t)=E\tau\mathrm{Sa}(\frac{t\tau}{2})\rightarrow F[R(t)]=2\pi E[u(\omega+\frac{\tau}{2})-u(\omega-\frac{\tau}{2})]$$

单选题 1分

设置

Given $\mathrm{F}[\mathrm{sgn}(t)] = \dfrac{2}{j\omega}$, **find** $\mathrm{F}^{-1}[\, j\pi\,\mathrm{sgn}(\omega)]$

A $\dfrac{1}{t}$

B $\dfrac{2\pi}{t}$

C $-\dfrac{2\pi}{t}$

D $-\dfrac{1}{t}$

提交

Example 3-2: **Calculate** $\mathcal{F}^{-1}[j\pi\,\mathrm{sgn}(\omega)]$.

Solution: $\because F(t) = j\pi\,\mathrm{sgn}(t)$

$$\mathcal{F}[F(t)] = j\pi\frac{2}{j\omega} = \frac{2\pi}{\omega}$$

$$\therefore \mathcal{F}^{-1}[j\pi\,\mathrm{sgn}(\omega)] = \frac{1}{2\pi}\mathcal{F}[F(t)]\Big|_{\omega=-t} = \frac{1}{2\pi}\left[\frac{2\pi}{\omega}\right]\Bigg|_{\omega=-t} = -\frac{1}{t}$$

$$j\pi\,\mathrm{sgn}(t) \leftrightarrow \frac{2\pi}{\omega}$$

$$\frac{-1}{2\pi}\left(\frac{2\pi}{t}\right) \leftrightarrow j\pi\,\mathrm{sgn}(\omega)$$

$$-\frac{1}{t} \leftrightarrow j\pi\,\mathrm{sgn}(\omega)$$

sgn(t) is an odd function.

3.7.3 Parity, Realness, and Imaginariness

$$F(\omega)=\int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt=\int_{-\infty}^{\infty} f(t)\cos(\omega t)dt-j\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt=R(\omega)+jX(\omega)$$

$$\begin{cases} R(\omega)=\int_{-\infty}^{\infty} f(t)\cos(\omega t)dt \\ X(\omega)=-\int_{-\infty}^{\infty} f(t)\sin(\omega t)dt \end{cases}$$

If $f(t)$ is a real-valued function, then:

- **The real part of $F(\omega)$ $\left(R(\omega)\right)$ is even;**
- **The imaginary part of $F(\omega)$ $\left(X(\omega)\right)$ is odd.**

Case 1: $f(t)$ is real and even $(f(-t)=f(t))$

$\sin(\omega t)$ is odd, so $X(\omega)=0$; $F(\omega)=R(\omega)$ **(real, even).**

Example: $f(t)=e^{-\alpha|t|}\rightarrow F(\omega)=\frac{2\alpha}{\alpha^2+\omega^2}$.

Case 2: $f(t)$ is real and odd $(f(-t)=-f(t))$

$\cos(\omega t)$ is even, so $R(\omega)=0$; $F(\omega)=jX(\omega)$ **(purely imaginary, odd).**

Example: $f(t)=\text{sgn}(t)\rightarrow F(\omega)=\frac{2}{j\omega}$.

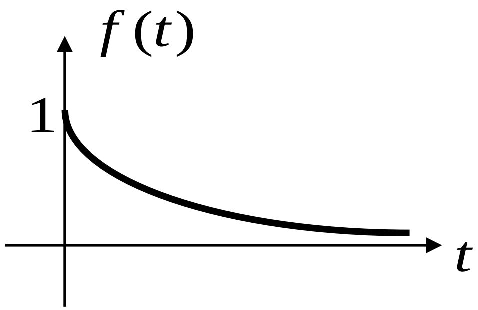
3.7 Basic Properties of Fourier Transform

$$F(\omega) = |F(\omega)| e^{j\varphi(\omega)} = R(\omega) + jX(\omega)$$

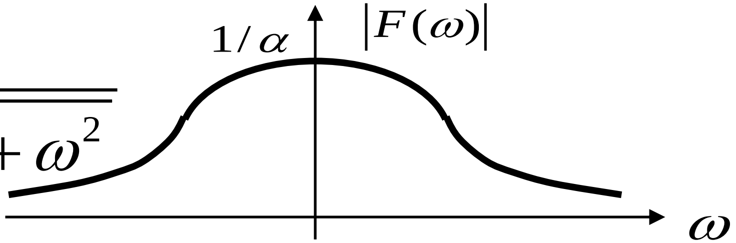
$$|F(\omega)| = \sqrt{R^2(\omega) + X^2(\omega)}, \quad \varphi(\omega) = \arctan \frac{X(\omega)}{R(\omega)}$$

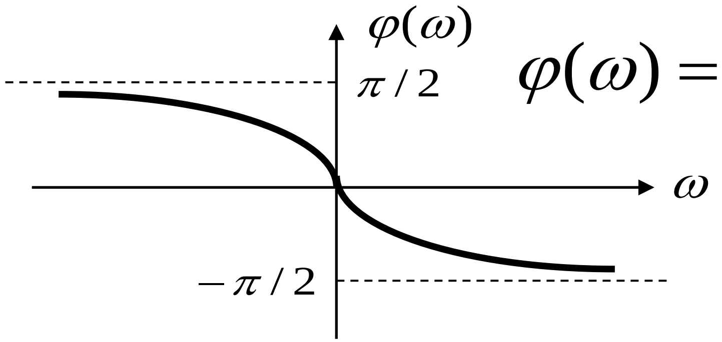
If f(t) is real or purely imaginary $(f(t) = jg(t),\ g(t)$ real), then $|F(\omega)|$ is even, and $\varphi(\omega)$ is odd.

Example: $f(t) = e^{-\alpha t} u(t)$



$$|F(\omega)| = \frac{1}{\sqrt{\alpha^2 + \omega^2}}$$





$$\varphi(\omega) = -\arctan(\frac{\omega}{\alpha})$$

3.7.4 Shift Property

(1) Time-Shift Property

If $F(\omega) = \mathcal{F}[f(t)]$

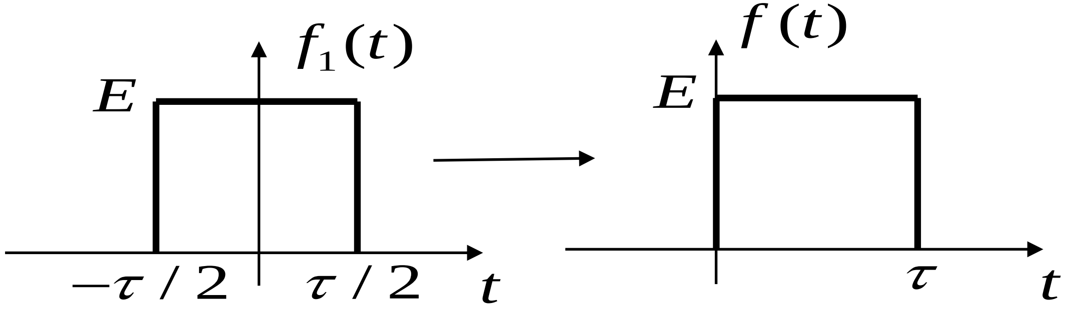
Then $\mathcal{F}[f(t-t_0)] = \int_{-\infty}^{\infty} f(t-t_0)e^{-j\omega t}dt = e^{-j\omega t_0}\int_{-\infty}^{\infty} f(t-t_0)e^{-j\omega(t-t_0)}d(t-t_0)$

$$= F(\omega)e^{-j\omega t_0} = |F(\omega)|e^{j\varphi(\omega)}e^{-j\omega t_0}$$

Amplitude spectrum remains unchanged.
Phase spectrum has a phase shift of -ωt_0
(proportional to frequency).

Similarly, $\mathcal{F}[f(t+t_0)] = F(\omega)e^{j\omega t_0}$

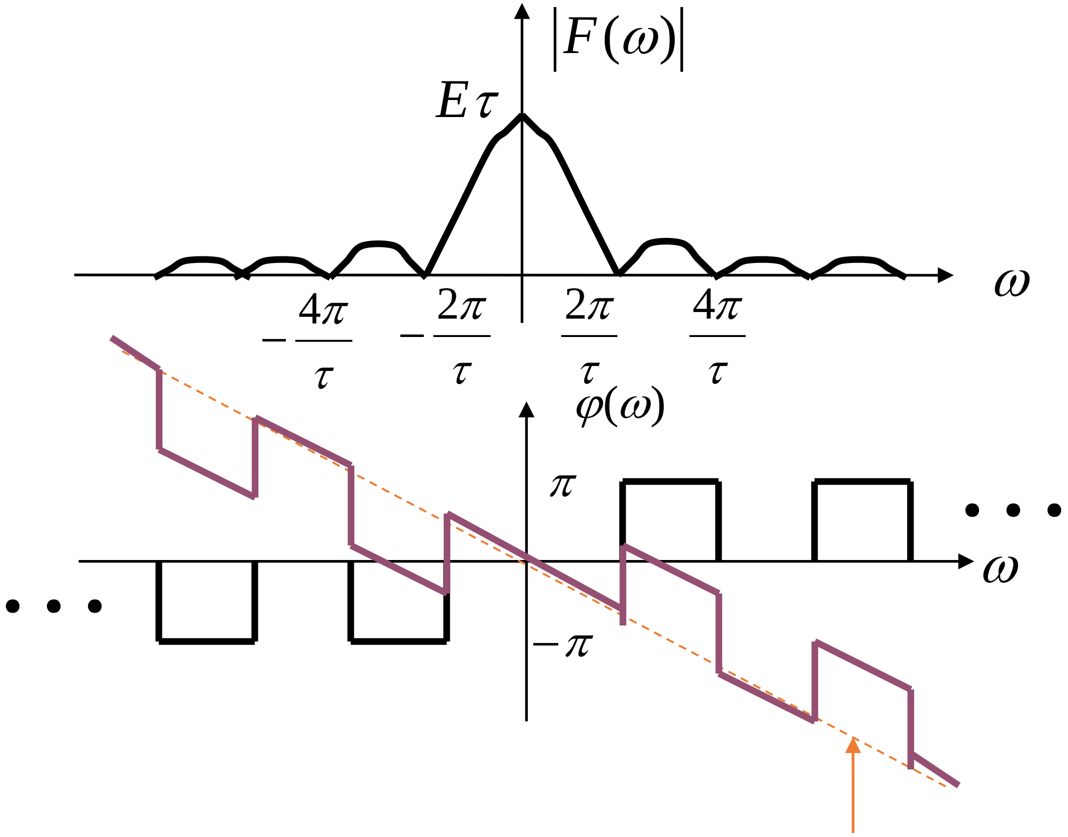
Example 3-3: **Find the spectral function of the single-sided rectangular pulse signal *f*(t) shown in the figure.**



Solution: *f*(t) is the right time shift of the symmetric gate function f_1(t) by τ/2.

$$F_1(\omega) = E\tau\,\mathrm{Sa}\,(\frac{\omega\tau}{2})$$

$$F(\omega) = E\tau\,\mathrm{Sa}\,(\frac{\omega\tau}{2})e^{-j\omega\frac{\tau}{2}}$$



The amplitutde spectrum remains unchanged. The phase spectrum has a phase shift of $-\omega\tau/2$.

单选题 1分

设置

Find the Fourier Transform of $x(t) = e^{-t+2}u(t\text{-}2)$

A $e^{-j2\omega}\dfrac{1}{1+j2\omega}$

B $e^{-j2\omega}\dfrac{1}{1+j\omega}$

C $e^{-j\omega}\dfrac{1}{1+j2\omega}$

D $2e^{-j\omega}\dfrac{1}{1+j\omega}$

提交

Solution:

Step 1: Fourier Transform of Basic Exponential Signal

Let $x_1(t) = e^{-t}u(t)$.Its Fourier Transform is:

$$\mathcal{F}[x_1(t)] = \frac{1}{1 + j\omega}$$

Step 2: Express $x(t)$ as Time-Shifted $x_1(t)$

$$x(t) = e^{-(t-2)}u(t-2) = x_1(t-2)$$

(Shift $x_1(t)$to the right by $t_0 = 2$.)

Step 3: Apply Time-Shift Property

$$\mathcal{F}[x(t)] = \mathcal{F}[x_1(t-2)] = \mathcal{F}[x_1(t)] \cdot e^{-j\omega \cdot 2} = \frac{e^{-j2\omega}}{1 + j\omega}$$

(2) Frequency-Shift Property (Modulation Principle)

If $\mathcal{F}[f(t)] = F(\omega)$, then:

$$\mathcal{F}\left[f(t)e^{j\omega_0 t}\right] = F(\omega - \omega_0)$$
$$\mathcal{F}\left[f(t)e^{-j\omega_0 t}\right] = F(\omega + \omega_0)$$

Proof:

$$\mathcal{F}\left[f(t)e^{j\omega_0 t}\right] = \int_{-\infty}^{+\infty} f(t)e^{j\omega_0 t} \cdot e^{-j\omega t} dt = \int_{-\infty}^{+\infty} f(t)e^{-j(\omega-\omega_0)t} dt = F(\omega - \omega_0)$$

Example 3-4: Find the Fourier transforms of $e^{j\omega_0 t}$, $\cos\omega_0 t$ and $\sin\omega_0 t$ by the frequency-shift property.

Solution: $\mathrm{F}[1] = 2\pi\delta(\omega)$ $\qquad$ $\mathrm{F}[e^{j\omega_0 t}] = 2\pi\delta(\omega-\omega_0)$

$$\because \cos\omega_0 t = \frac{1}{2}[e^{j\omega_0 t} + e^{-j\omega_0 t}] \qquad \therefore \mathrm{F}[\cos\omega_0 t] = \pi[\delta(\omega+\omega_0) + \delta(\omega-\omega_0)]$$

$$\because \sin\omega_0 t = \frac{1}{2j}[e^{j\omega_0 t} - e^{-j\omega_0 t}] \qquad \therefore \mathrm{F}[\sin\omega_0 t] = j\pi[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)]$$

Modulation (Key Application)

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$ $\qquad$ $\mathcal{F}[f(t)e^{-j\omega_0 t}] = F(\omega + \omega_0)$

$$\because \cos\omega_0 t = \frac{1}{2}[e^{j\omega_0 t} + e^{-j\omega_0 t}] \qquad \sin\omega_0 t = \frac{1}{2j}[e^{j\omega_0 t} - e^{-j\omega_0 t}]$$

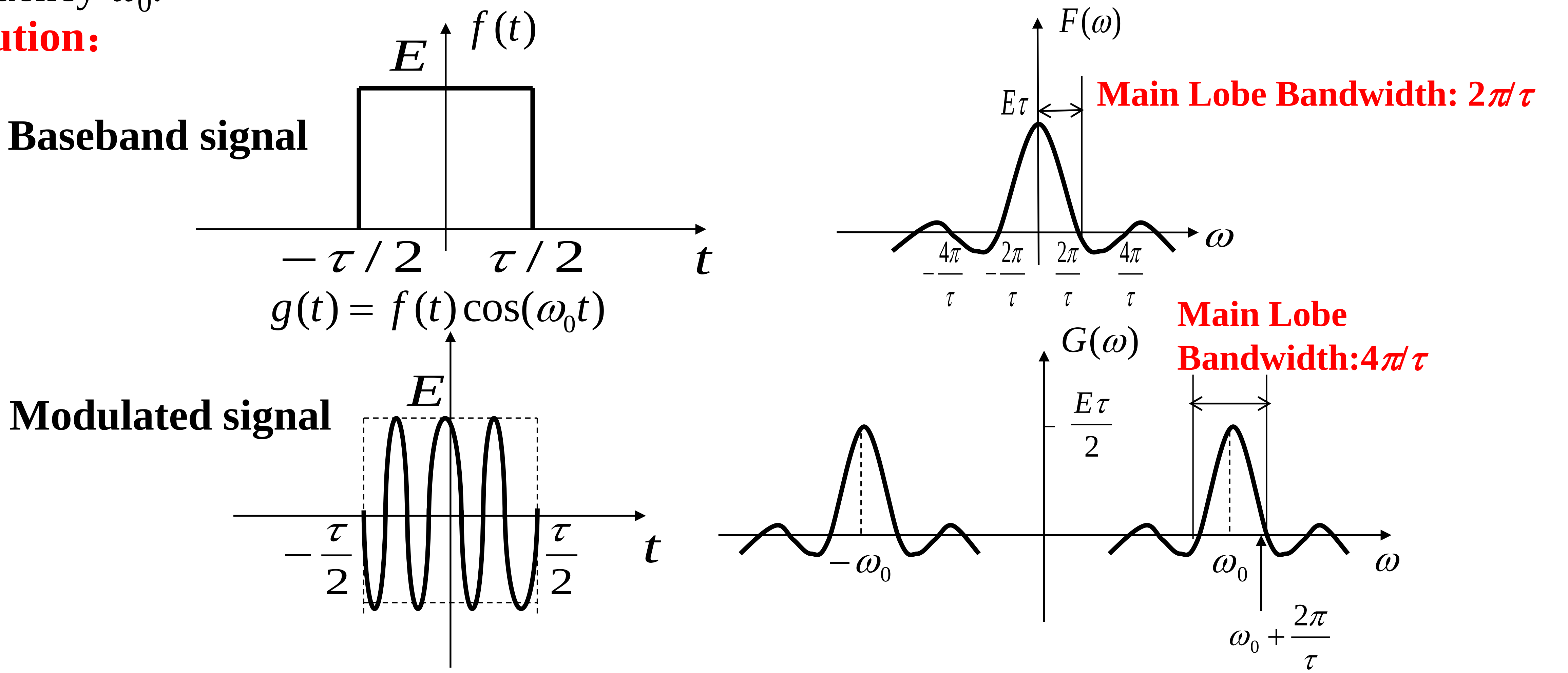
$$\therefore \mathcal{F}[f(t)\cos\omega_0 t] = \frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)]$$

$$\mathcal{F}[f(t)\sin\omega_0 t] = \frac{j}{2}[F(\omega + \omega_0) - F(\omega - \omega_0)]$$

Physical Meaning: The spectrum of $f(t)$ is shifted to $\pm\omega_0$ (**modulation shifts the baseband spectrum to the carrier frequency band**).

Example 3-5： Find the Fourier transform of a rectangular pulse Amplitude Shift Keying (ASK) signal and draw the spectrum, with pulse amplitude E, pulse width τ and carrier frequency ω_0.

Solution：



$$F(\omega) = E\tau \mathrm{Sa}(\frac{\omega\tau}{2})$$

$$G(\omega) = \frac{1}{2}[F(\omega+\omega_0) + F(\omega-\omega_0)]$$

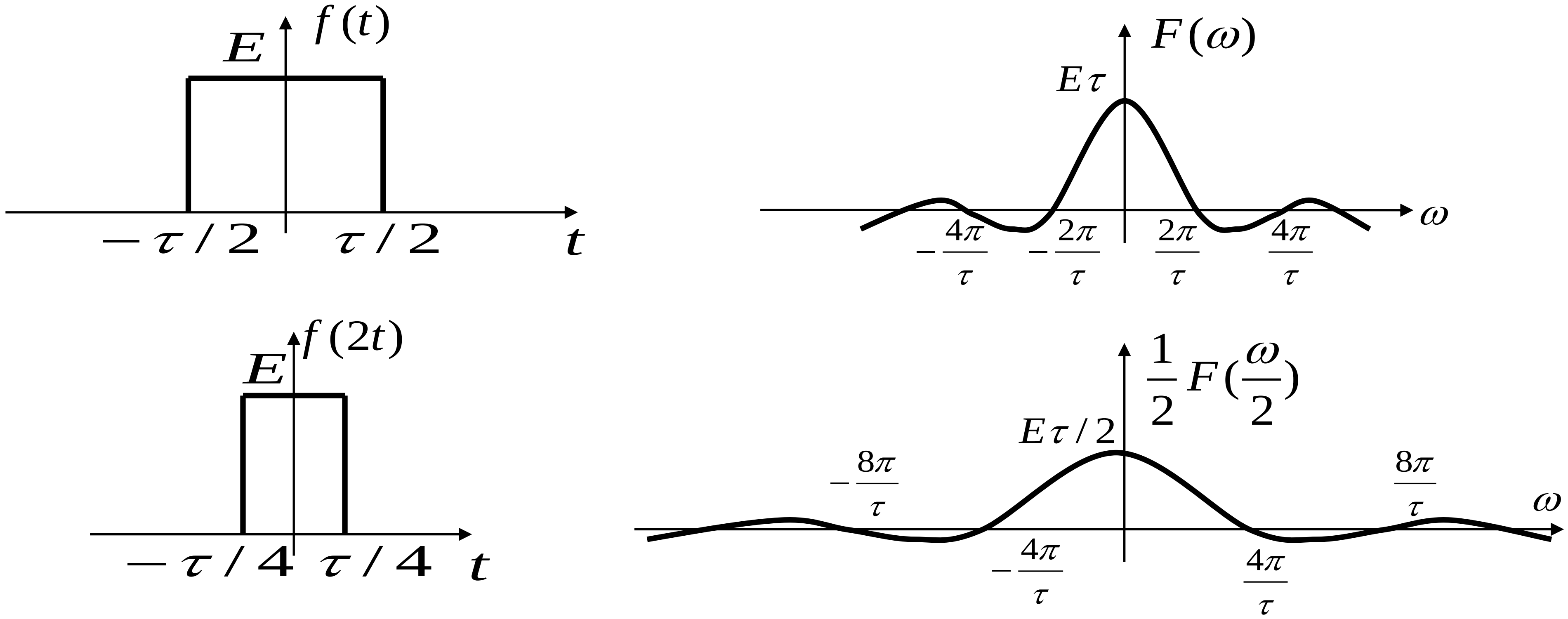
$$= \frac{E\tau}{2}\{\mathrm{Sa}[(\omega+\omega_0)\frac{\tau}{2}] + \mathrm{Sa}[(\omega-\omega_0)\frac{\tau}{2}]\}$$

Conclusion: **During modulation, (1) bandwidth doubles; (2) spectral density halves.**

3.7.5 Scaling Property

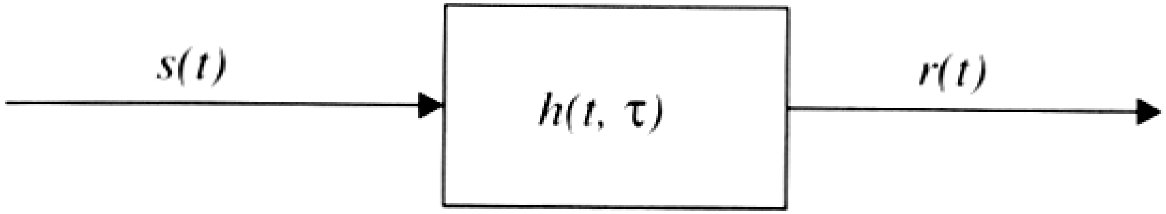
If $F(\omega) = \mathcal{F}[f(t)]$ **Then** $\mathcal{F}[f(at)] = \dfrac{1}{|a|} F(\dfrac{\omega}{a})$

Compressing a signal in time domain by *a* (*a*>1) times (signal changes faster) is equivalent to expanding the spectrum by *a* times in frequency domain (more high-frequency components) and reducing the spectral amplitude by 1/*a* (energy conservation). Similarly, time expansion is equivalent to frequency compression.



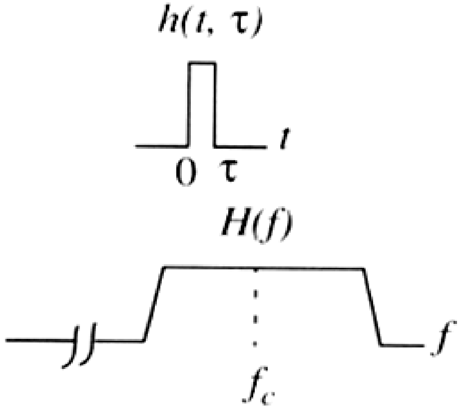
3.7 Basic Properties of Fourier Transform

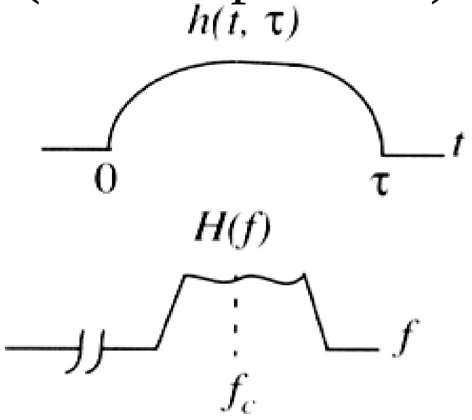
Example: Time Expansion and Frequency Compression of Communication Channels



Small channel delay

larger channel delay
(time expansion)





Large channel bandwidth

Smaller bandwidth
(frequency compression)

Special case: $\mathcal{F}[f(-t)] = F(-\omega)$ （**time reversal, *a* = -1**）

Combined Property: Scaling + Time Shift

$$\boxed{\mathcal{F}[f(at-t_0)] = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)e^{-j\frac{\omega t_0}{a}}}$$

Proof:

$$f(t) \leftrightarrow F(\omega)$$

Scaling $$f(at) \leftrightarrow \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$$

Time shift $$f\left[a\left(t-\frac{t_0}{a}\right)\right] \leftrightarrow \frac{1}{|a|}F\left(\frac{\omega}{a}\right)e^{-j\frac{\omega t_0}{a}}$$

$$f(t) \leftrightarrow F(\omega)$$

Time shift $$f(t-t_0) \leftrightarrow F(\omega)e^{-j\omega t_0}$$

Scaling $$f(at-t_0) \leftrightarrow \frac{1}{|a|}F\left(\frac{\omega}{a}\right)e^{-j\frac{\omega t_0}{a}}$$

3.7.6 Differentiation and Integration Properties

(1) Time-Domain Differentiation Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}\left[\frac{df(t)}{dt}\right] = j\omega F(\omega)$, $\quad \mathcal{F}\left[\frac{d^n f(t)}{dt^n}\right] = (j\omega)^n F(\omega)$

Proof: $$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega$$

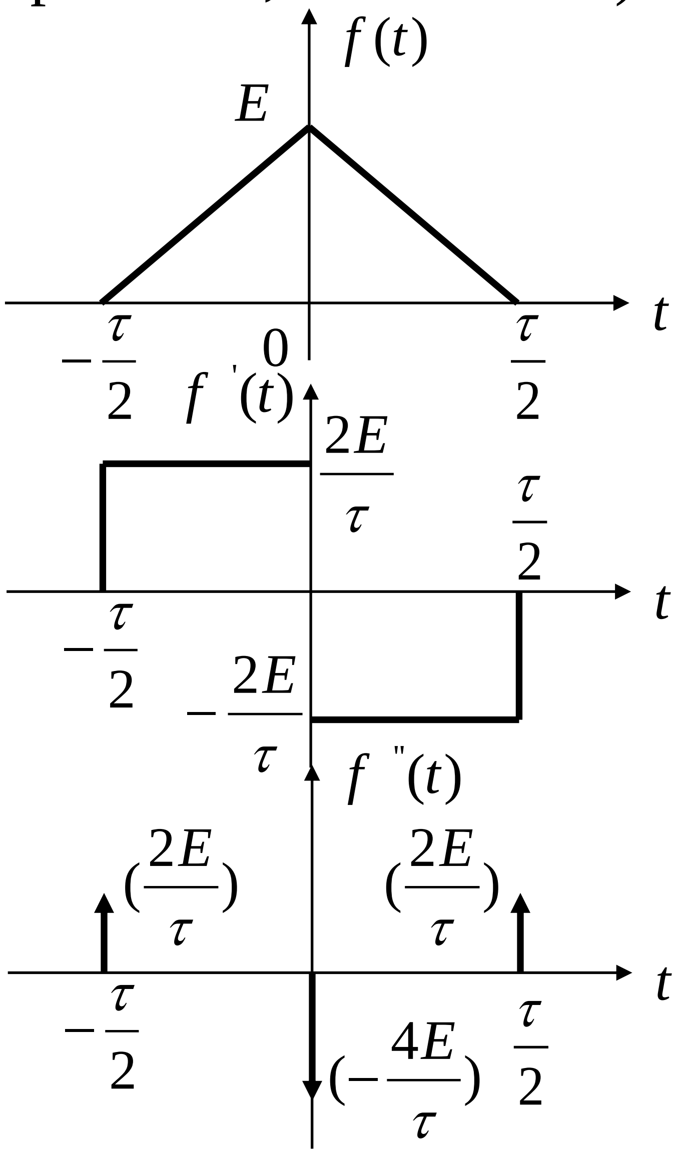
Differentiate both sides with respect to t:

$$\frac{d}{dt}f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)(j\omega)e^{j\omega t}d\omega$$

$$\mathcal{F}[f'(t)] = j\omega F(\omega)$$

Example: $\mathcal{F}[\delta(t)] = 1$ $\quad \mathcal{F}[\delta'(t)] = j\omega$ $\quad \mathcal{F}[\delta^{(n)}(t)] = (j\omega)^n$

Example 3-6: Find the Fourier Transform of the triangular pulse *f*(t) shown in the figure (amplitude *E*, base width τ).



Solution:

$$f_1(t) = f'(t) = \begin{cases} \frac{2E}{\tau} & -\frac{\tau}{2} < t < 0 \\ -\frac{2E}{\tau} & 0 < t < \frac{\tau}{2} \\ 0 & |t| > \frac{\tau}{2} \end{cases}$$

$$f_2(t) = f''(t) = \frac{2E}{\tau}[\delta(t+\frac{\tau}{2}) + \delta(t-\frac{\tau}{2}) - 2\delta(t)]$$

Take Fourier transform of both sides,

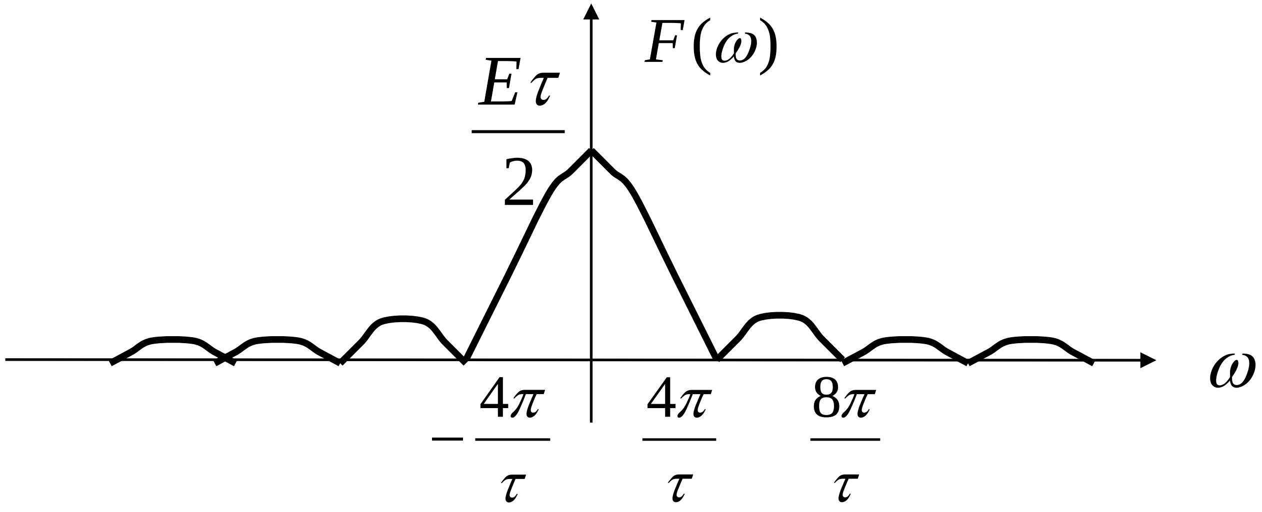
$$F_2(\omega) = \frac{2E}{\tau}(e^{j\omega\frac{\tau}{2}} + e^{-j\omega\frac{\tau}{2}} - 2) = \frac{4E}{\tau}\left[\cos\left(\omega\frac{\tau}{2}\right) - 1\right]$$

$$= -\frac{\omega^2 E\tau}{2}\mathrm{Sa}^2(\frac{\omega\tau}{4})$$

3.7 Basic Properties of Fourier Transform

$$F_2(\omega) = (j\omega)^2 F(\omega) = -\frac{\omega^2 E\tau}{2}\mathrm{Sa}^2(\frac{\omega\tau}{4})$$

$$\therefore \quad F(\omega) = \frac{E\tau}{2}\mathrm{Sa}^2(\frac{\omega\tau}{4})$$



(2) Time-Domain Integration Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $$\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)d\tau\right] = \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$$

where $F(0) = F(\omega)\big|_{\omega=0}$ **DC Component**

If $F(0) = 0$ **no DC Component**

Then $$\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)d\tau\right] = \frac{F(\omega)}{j\omega}$$

(3) Frequency-Domain Differentiation Property

If $F(\omega) = \mathcal{F}[f(t)]$

Then $\mathcal{F}[(-jt)f(t)] = \dfrac{dF(\omega)}{d\omega}$,

$$\mathcal{F}[(-jt)^n f(t)] = \frac{d^n F(\omega)}{d\omega^n}$$

$$\boxed{\mathcal{F}[t^n f(t)] = j^n \frac{d^n F(\omega)}{d\omega^n}}$$

Example: $\because \ \mathcal{F}[1] = 2\pi\delta(\omega)$

$\therefore \ \mathcal{F}[t] = j2\pi\delta'(\omega)$

$\mathcal{F}[t^n] = 2\pi j^n \delta^{(n)}(\omega)$

(4) Frequency-Domain Integration Property

If $\quad F(\omega) = \mathbf{F}[f(t)]$

Then $\quad \mathsf{F}^{-1}[\int_{-\infty}^{\omega} F(u)du] = \dfrac{f(t)}{-jt} + \pi f(0)\delta(t)$

If $\quad f(0) = 0$

Then $\quad \mathsf{F}^{-1}[\int_{-\infty}^{\omega} F(u)du] = \dfrac{f(t)}{-jt}$

Homework

- Repeat the examples in the lecture notes.
- 3.58(a)(b) and 3.62(c) in the textbook.